

3. Project Design Phase

Entity Relationship (ER) Diagram

```
USER ||--o{ DEBT_RECORD : has
```

USER

```
id (PK)
name
email
hashed_password
total_debt
```

DEBT_RECORD

```
id (PK)
loan_type
bank_name
loan_amount
interest_rate
emi_amount
overdue_months
debt_stress_level
ai_strategy
owner_id (FK -> USER.id)
```

Note: A rendered version of this diagram (in Mermaid format) is available in the GitHub repository README, where GitHub automatically renders it visually.

Use Case Diagram

Actor: **User**

Use Cases: Register Account, Login, Add Loan/Debt Record, View Dashboard, Get AI Negotiation Strategy, Get Settlement Prediction, Generate Negotiation Email, View Loan History, Delete Loan Record, Update Profile.

The User interacts directly with all the use cases listed above. “Add Loan” includes “Get AI Negotiation Strategy” and “Get Settlement Prediction” as part of its flow.

System Architecture

```
React Frontend  --(REST API calls)-->  FastAPI Backend
FastAPI Backend --(SQLAlchemy ORM)-->   SQLite Database
FastAPI Backend --(API calls)-->        Google Gemini AI
FastAPI Backend --(JWT)-->              Auth Layer
```